Lecturer's remarks
Dosente kommentaar

Brightlight Tutorials [Union, filtering and Aggregates

* SQL Joins

① Inner Joins

```
Select  Employee_id,
        First_name,
        department,
        Salary,
        project_id,
        project_name,
        Budget,
        Status
FROM  employees
Inner Join Projects
  ON  employee_id = employee_id;
```

Employee ID	First Name	Last Name	Depart	Salary	Project ID	Project name	Budget	Status
1	John	Doe	IT	70000	101	AI Develop	100000	Completed
2	Alice	Smith	HR	60000	102	Employee training	50000	ongoing
1	John	Doe	IT	70000	103	CyberSec Audit	75000	Pending
3	Bob	Johnson	Finace	75000	104	Financial Analysis	90000	ongoing
5	Emma	Wilson	Sales	65000	105	Market Expansion	65000	Completed
6	Michael	Clark	Finance	80000	106	Risk Manager	80000	Pending

② Left Join

```
Select  employee_id,
        first_name,
        Last_name,
        department,
        Salary,
        Project_id,
        Project_name,
        budget,
        Status
from  employees
Left Join Projects
  ON  employee_id = employee_id;
```

Employee ID	First name	Last name	Depart	Salary	Project ID	Project name
4	David	Brown	IT	72000	null	null

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③ Right Join

```
Select Project_ID,
       Project_name,
       Budget,
       Status,
       Employee_ID,
       First_name,
       Department,
       Salary;
From Employee
Right Join Projects
on employee_ID = Employee_ID;
```

[Same as Inner Join because all projects have Employees]

④ Full outer Join

```
Select employee_ID,
       First_name,
       Last_name,
       Department,
       Salary,
       Project_ID,
       Project_name
       budget,
       Status
From Employee
Full Outer Join Project
On employee_ID = employee_ID;
```

[Same as Left Join here Since no orphan projects]

* 2 Union

```
=> Union (unique)
Select city AS From employees
Union
Select Status AS Location From Projects;
```

Location	Complet
NY	ongoing
LA	Pendin
Toronto	Ongoin
London	Complet
Sydney	Pendu
~~Completed~~	

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⑥ Union All (with duplicates)

```
Select City AS Location
from employees
Union All
Select Status AS Location
from Project;
```

Location	
NY	Completed
LA	ongoing
Toronto	Pending
London	Ongoing
Sydney	Completed
NY	Pending

* Filtering

* ① Salary > 70000

```
Select employee_id, first_name, department, Salary
From employees
Where Salary > 70000;
```

Employee ID	first name	Last name	Department	Salary
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000
6	Michael	Clark	Finace	80000

② IT or Finance

```
Select employee_ID, First_name, Last_name, Department, Salary
From employees
Where Department = 'IT' or department = 'Finance';
```

Employee ID	First name	Last name	department	Salary
1	John	Doe	IT	7000
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000
6	Micheal	Clark	Finance	80000

③ Project Not completed

```
Select project_ID, Project_name, Budget, Status.
From Project
Where Status <> 'Completed';
```

Project ID	Project name	Budget	Status
102	Employee training	50000	ongoing
103	Cybersecurity Audit	75000	Pending
104	Financial Analysis	90000	ongoing
106	Risk management	80000	Pending

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(10) Budget > 70,000 and not Completed

```
Select Project_ID, Project_name, budget, Status
from Projects
Where Budget > 70000
And Status <> 'Completed';
```

Project ID	Project name	Budget	Status
103	Cybersecurit Audit	75000	Pending
104	Financial Analysis	90000	ongoing
106	Risk Management	80000	Pending

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(11) Employees from NY or toronto (DESC Salary)

```
Select employee_id, first_name, lastname, department, salary, city
from employees
Where City = 'New York' OR City = 'toronto'
Order By Salary DESC;
```

Employee ID	First name	Last name	Department	Salary	City
6	Micheal	Clark	Finace	80000	NY
3	Bob	Johnson	Finace	75000	Toronto
1	John	Doe	IT	70000	NY

(12) Top 3 highest paid

```
Select employee_id, first_name, department, Salary
from employees
Order By Salary DESC
Limit 3;
```

Employee ID	First name	Last name	Department	Salary
6	Micheal	Clark	Finance	80000
3	Bob	Johnson	Finace	75000
4	David	Brown	IT	72000

* Aggregate

(13) total Salary per department

```
Select department,
       Sum(Salary) AS total_Salary
from employees
Group By Department
Order By total_Salary DESC;
```

Department	total Salary
Finance	155000
IT	142000
Sales	65000
HR	60000

(14) AVG Salary per city (>65,000)

```
Select City,
       AVG(Salary) AS Average_Salary
from employees
Group By City
Having AVG(Salary) >65000;
```

City	Average Salary
NY	75000
toronto	75000
London	72000
Sydney	65000

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⑮ Employees per department (>1)

```
Select department
       Count (employee_id) AS employee_count
From employees
Group By department
Having Count (employee_ID) >1;
```

Department	Employee Count
IT	2
Finance	2

⑯ Project per Status (≥2)

```
Select Status,
       Count (Project_id) AS Project_Count
From Projects
Group By Status
Having Count (Project_id) >=2;
```

Status	Project Count
Complete	2
ongoing	2
Pending	2

⑰ Total project budget per Employee (>150,000)

```
Select employee_id, First_name, Last_name,
       Sum (Budget) AS total_Project_budget
From employees
Join Projects
On employee_id = employee_id
Group By employee_id, First_name, last_name
Having Sum (Budget) > 150000;
```

Employee ID	First Name	Last Name	total Project Budget
1	John	Doe	175000

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